

The background of the slide is a composite image. The top half shows a large pile of elephant tusks of various sizes and shapes, some with handwritten numbers, laid out on a dark surface. The bottom half shows a close-up of an elephant's head, focusing on its eye and the base of its tusk.

SNP Panel Development for Conservation Applications

Advanced Conservation Genomics —
DAY 3

How this session is structured

- 1 Why marker panels matter in conservation**
When they are used and real-world case studies

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- 5 Cost-efficiency & scalability**
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Cost-efficiency & scalability

Choosing a platform for a real monitoring programme

Case study



1. Why Marker Panels Matter in Conservation

- Where panels are used today
- Real-world case studies
- Different markers

In conservation it matters if we reach a decision in time

Nowadays, we can generate genome-wide data for thousand of species — but most on-the-ground decisions cannot wait for, or afford, a new WGS run every time.

Prosecution

Needs a result before a court deadline

Hatchery

Needs to know which salmon run arrived this week

Breeding programme

Needs a kinship check before pairing two individuals, every season

Field work

Needs a species/hybrid call from a sample, cheaply, at scale

THE ROLE OF A MARKER PANEL

A targeted panel is the bridge: the same statistical logic as a WGS analysis, compressed into an assay that is fast, cheap, standardised and deployable well outside a genomics lab.

Where panels are used in conservation today

1

Wildlife Forensics & Anti-Poaching

Tracing ivory, horn, bushmeat and other products to a geographic or population origin to support enforcement and prosecution.

2

Fisheries & Stock Management

Identifying which run or stock an individual belongs to, in-season, to manage harvest and protect at-risk populations.

3

Hybridisation & Introgression

Detecting hybrids and tracking gene flow between wild species, domestic relatives, or reintroduced and resident populations.

4

Population & Diversity Monitoring

Routine, repeatable tracking of genetic diversity and inbreeding in small or declining populations.

5

Captive Breeding & Reintroduction

Verifying pedigrees, minimising kinship in breeding pairs, and confirming the genetic success of releases.

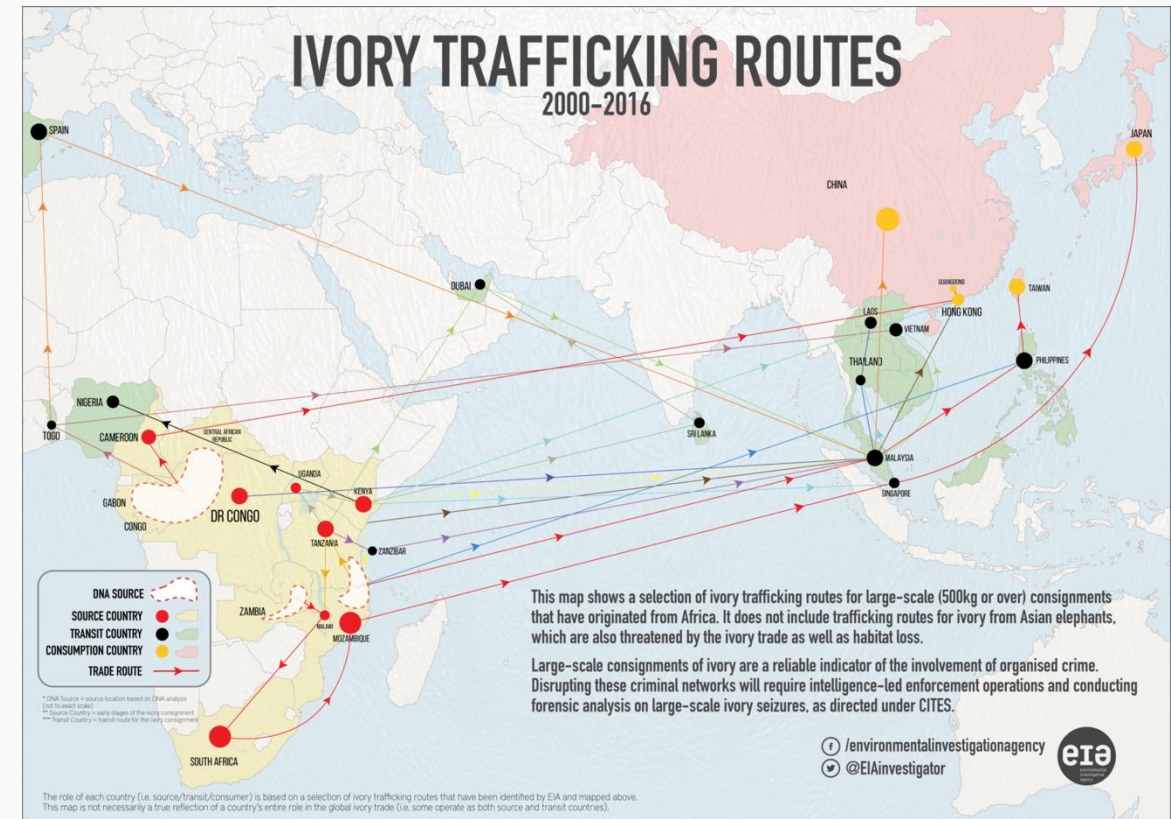
6

Trade & Biosecurity Compliance

Verifying the legal origin of regulated wildlife, fisheries or timber products entering trade.

Tracing poached ivory back to its source

- African elephants have been poached at extreme rates — an estimated 40,000 killed in 2011 alone — with large confiscations (≥ 0.5 tons) linked to organised transnational crime that is hard to police once ivory crosses borders.



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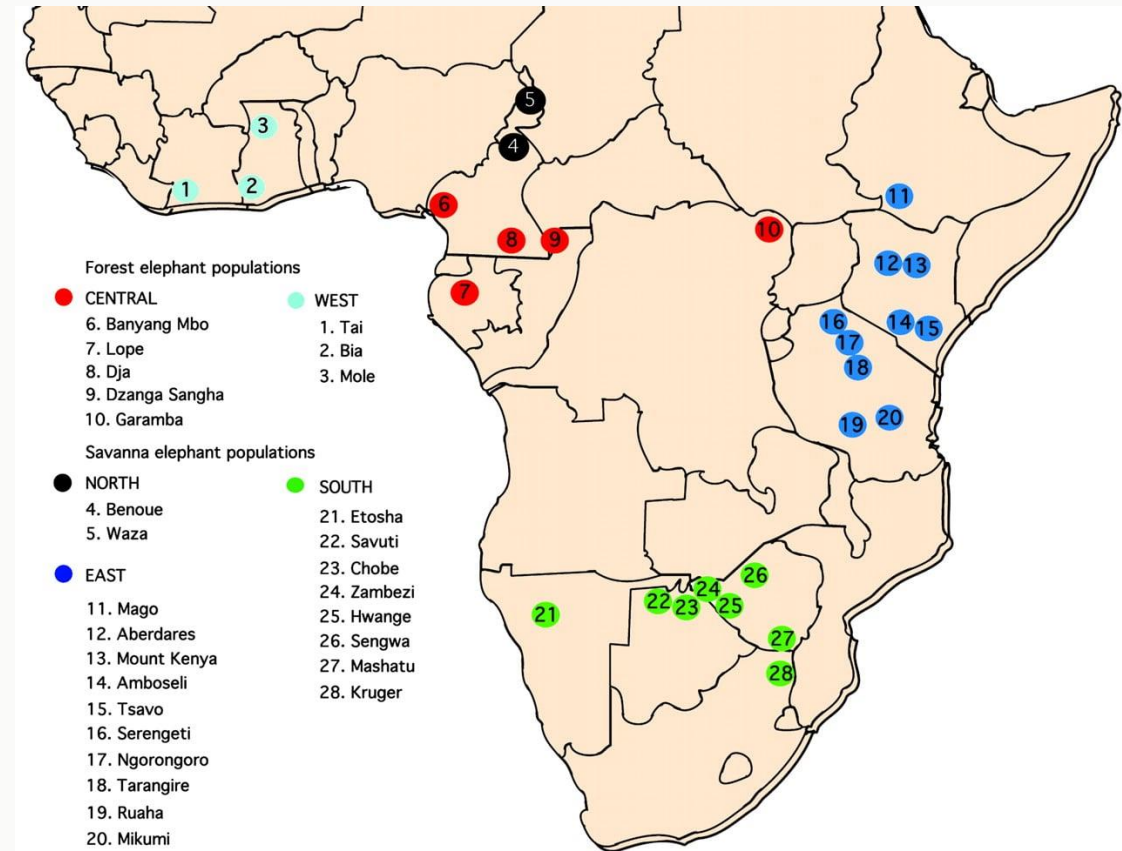
Assigning African elephant DNA to geographic region of origin: Applications to the ivory trade

Samuel K. Wasser^{*,†}, Andrew M. Shedlock^{*,†}, Kenine Comstock^{*,§}, Elaine A. Ostrander[§], Benezeth Mutayoba[¶], and Matthew Stephens[§]

^{*}Department of Biology, Center for Conservation Biology, University of Washington, Box 351800, Seattle, WA 98195; [†]Clinical Research and Human Biology Divisions, Fred Hutchinson Cancer Research Center, 1100 Fairview Avenue North, D4-100, Seattle, WA 98109-1024; [§]Department of Veterinary Physiology, Biochemistry, Pharmacology, and Toxicology, Sokoine University of Agriculture, P.O. Box 3017, Morogoro, Tanzania; and [¶]Department of Statistics, University of Washington, Seattle, WA 98195

Edited by Henry C. Harpending, University of Utah, Salt Lake City, UT, and approved August 24, 2004 (received for review May 7, 2004)

Map of Africa showing the collection sites



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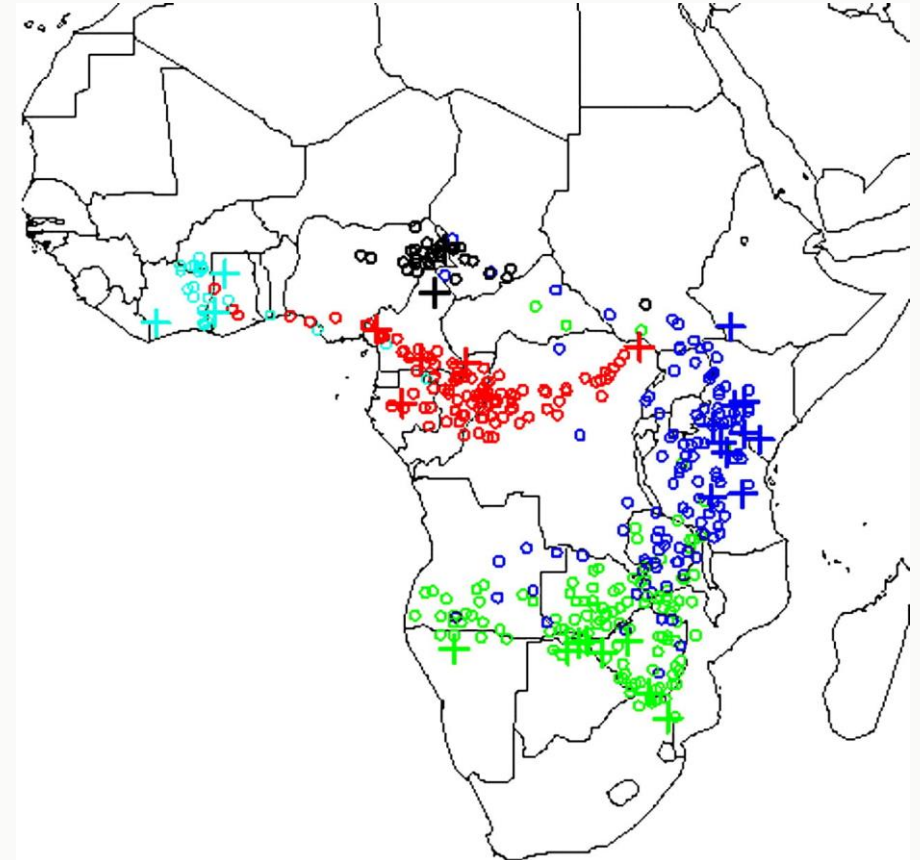
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Estimated locations of elephant tissue and fecal samples from across Africa



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Genetic assignment of large seizures of elephant ivory reveals Africa's major poaching hotspots

S. K. Wasser,^{1*} L. Brown,² C. Mailand,¹ S. Mondol,^{1†} W. Clark,³ C. Laurie,² B. S. Weir²

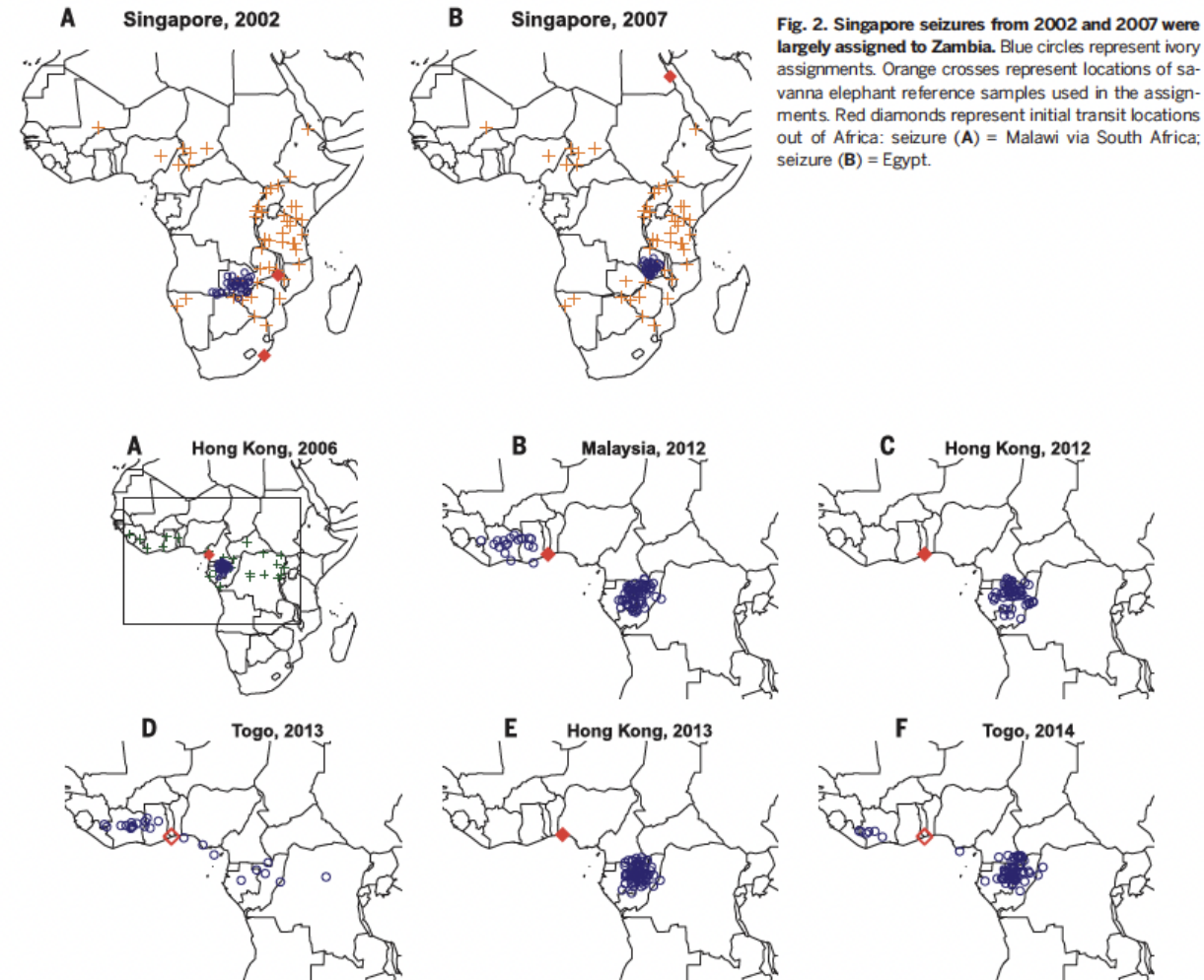


Fig. 3. Forest elephant seizures from 2006 to 2014 were largely assigned to the TRIDOM in northeastern Gabon, northwestern Republic of Congo, and southeastern Cameroon and neighboring Dzanga Sanga in southwestern CAR. Forest elephant reference sample locations (green crosses) are shown in the first map only, along with a box designating the magnified areas displayed in all subsequent maps. Blue circles represent ivory assignments. Initial transit locations out of Africa (solid diamonds) and areas where ivory was seized prior to transit out of Africa (open diamonds) include the following: seizure (A) = Cameroon; seizures (B) to (D) and (F) = Togo; seizure (E) = Nigeria.

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The pioneering work used microsatellite loci; the field is now moving toward SNP-based panels for lower DNA input and easier cross-lab standardisation.

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Research paper

Mini-SNaPshot multiplex assays authenticate elephant ivory and simultaneously identify the species origin

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DOI: 10.1002/ece3.3854

ORIGINAL RESEARCH

WILEY Ecology and Evolution

Single-nucleotide polymorphism discovery and panel characterization in the African forest elephant

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Telling salmon runs apart, in season

- Chinook salmon runs (fall / late-fall / winter / spring) are managed separately under the US Endangered Species Act, but runs are indistinguishable by morphology and mix together during migration.

Received: 10 February 2016 | Revised: 18 August 2016 | Accepted: 19 August 2016

DOI: 10.1002/ece3.2493

ORIGINAL RESEARCH

WILEY *Ecology and Evolution* Open Access

Sequencing improves our ability to study threatened migratory species: Genetic population assignment in California's Central Valley Chinook salmon

Mariah H. Meek^{1,2}  | Melinda R. Baerwald² | Molly R. Stephens^{2,3} | Alisha Goodbla² | Michael R. Miller² | Katharine M. H. Tomalty² | Bernie May²



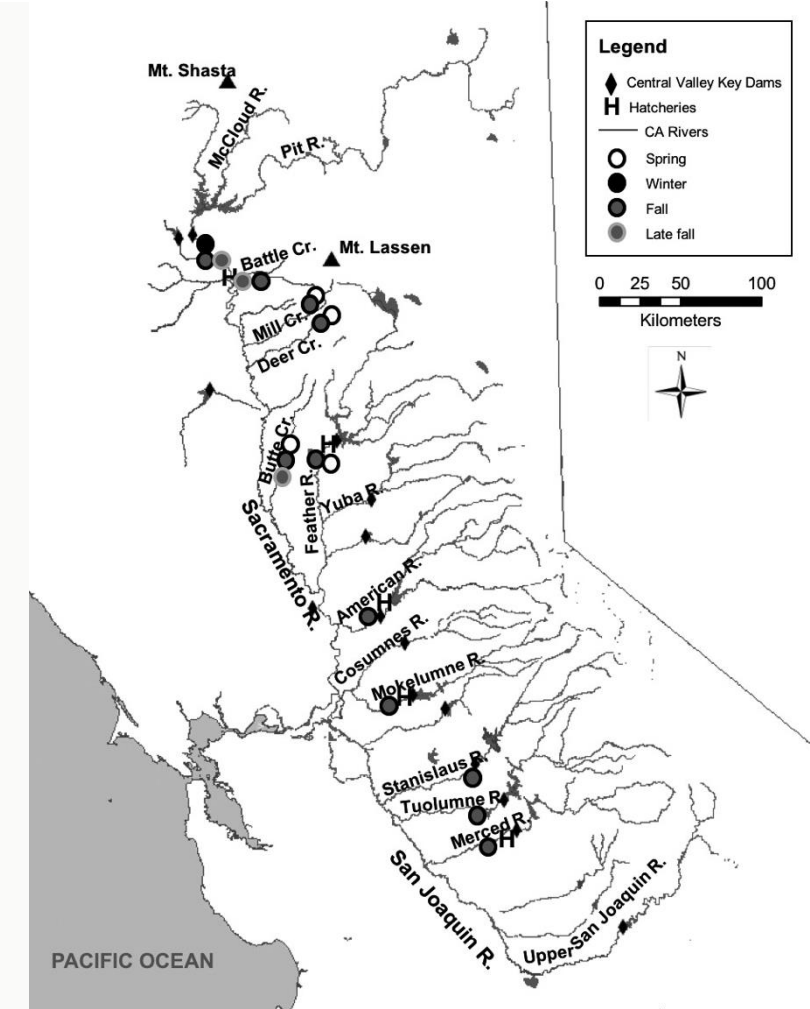
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- Chinook salmon runs (fall / late-fall / winter / spring) are managed separately under the US Endangered Species Act, but runs are indistinguishable by morphology and mix together during migration.
- They identify single-nucleotide polymorphisms (SNPs) that allow **fast and accurate identification of the imperiled Chinook salmon** in the Great Central Valley of California.
- They show that 80 well-chosen loci, drawn from a pool of over 11,500 SNPs developed from restriction site-associated DNA sequencing, can **accurately identify Chinook salmon runs and select populations within run.**

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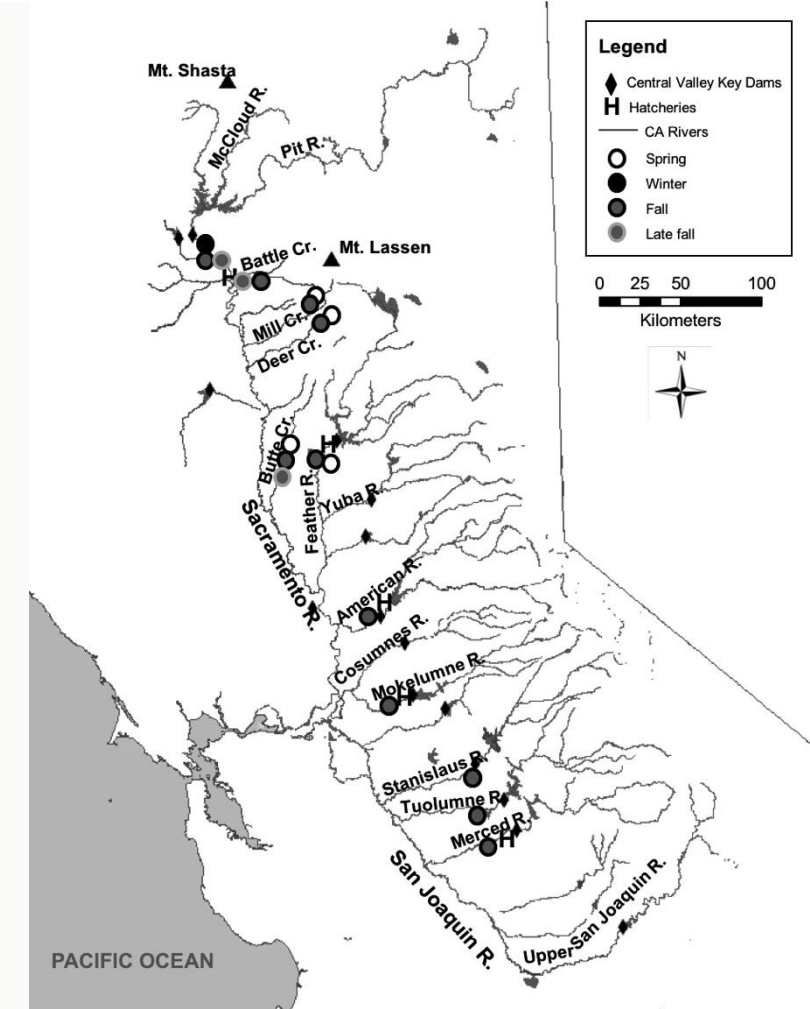
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This panel will allow to fine-tune the ecological understanding of the adaptive differences among the unique populations of Chinook salmon in the Central Valley and target restoration activities to aid those populations that are most imperiled by the effects of climate change.

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One species, multiple panels and goals



One species, multiple panels and goals

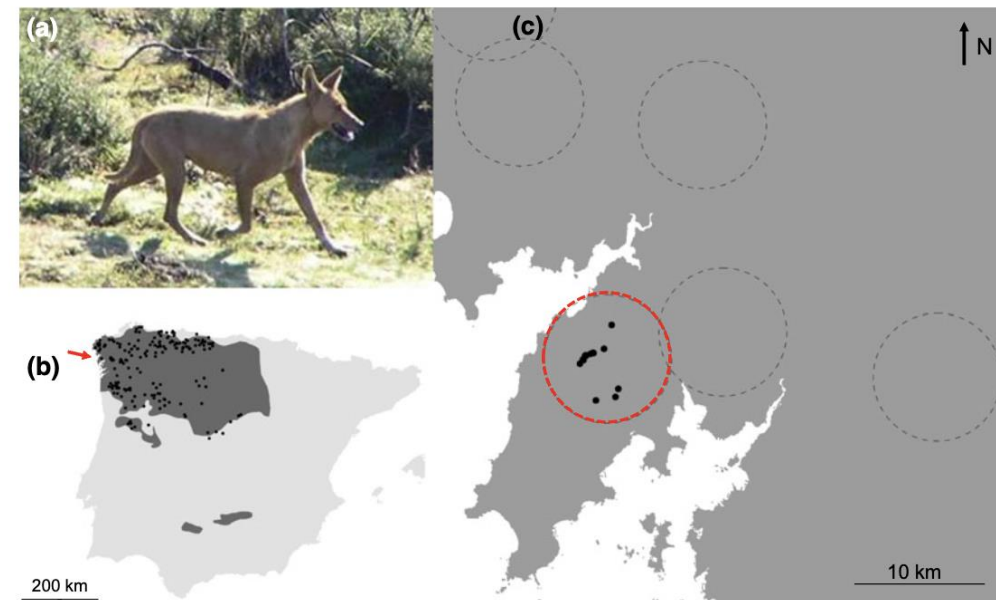
- Wolves and dogs provide a paradigmatic example of the ecological and conservation implications of hybridization events between wild and domesticated forms.

MOLECULAR ECOLOGY
RESOURCES
Molecular Ecology Resources (2014) doi: 10.1111/1755-0998.12313

Real-time assessment of hybridization between wolves and dogs: combining noninvasive samples with ancestry informative markers

RAQUEL GODINHO,* JOSÉ VICENTE LÓPEZ-BAO,†‡ DIANA CASTRO,* LUÍS LLANEZA,*§ SUSANA LOPES,* PEDRO SILVA*¶ and NUNO FERRAND*¶

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One species, multiple panels and goals

- Wolves and dogs provide a paradigmatic example of the ecological and conservation implications of hybridization events between wild and domesticated forms.
- They demonstrate that the 13 informative markers are as accurate as the 52 markers that were chosen without regard to the power to differentiate between wolves and dogs, also having the advantage of being rapidly screened on noninvasive samples. The efficiency of this STR panel significantly outperformed ten random sets of similar size and an additional commercial set of 18 markers.

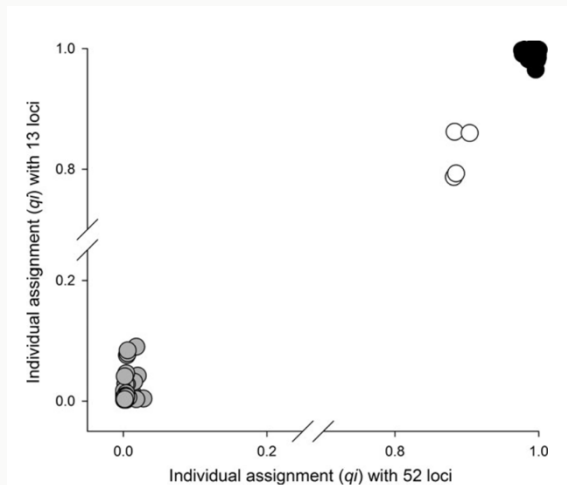
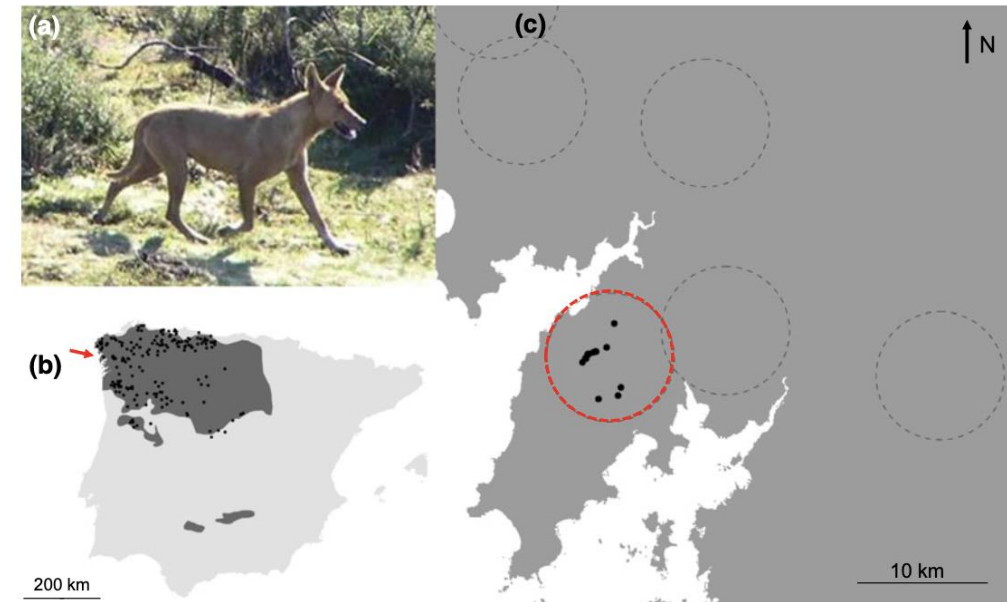


Fig. 3 Relationship between the individual probabilistic assignment (q_i) with 52 and 13 loci for Iberian wolves (black), dogs (grey) and wolf-dog hybrids (white).

Real-time assessment of hybridization between wolves and dogs: combining noninvasive samples with ancestry informative markers

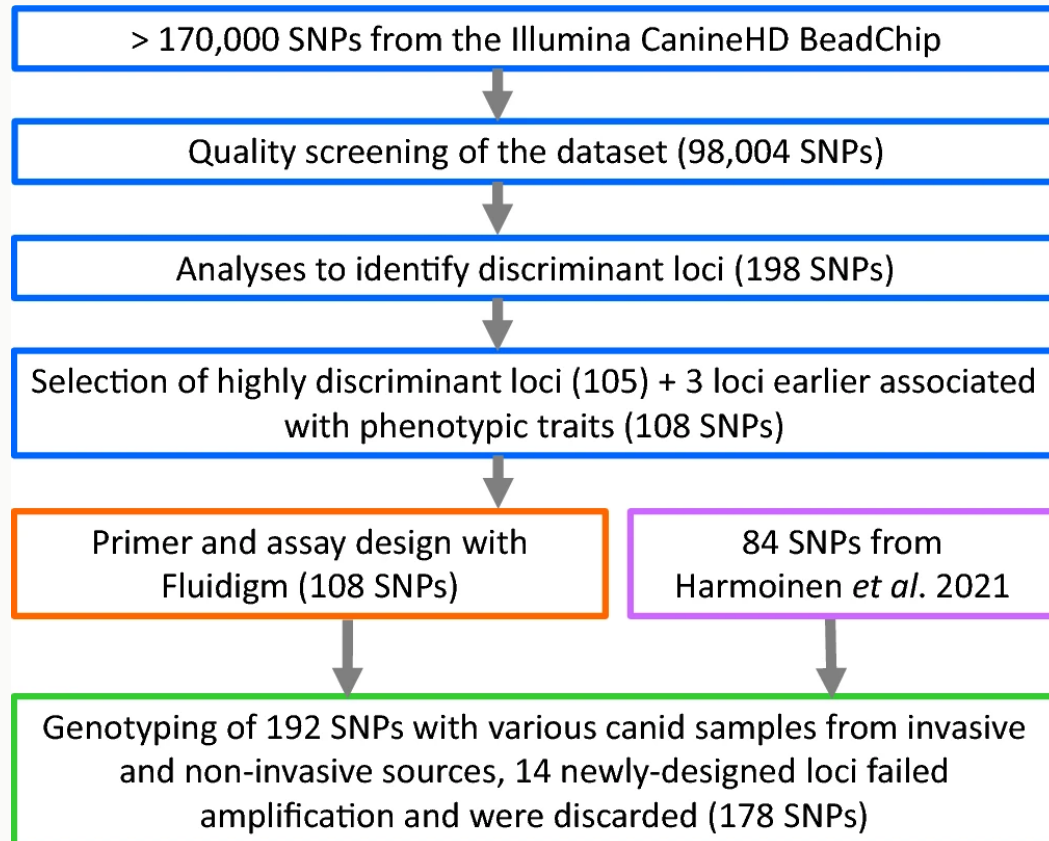
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This provides managers with a reliable tool to evaluate hybridization and estimate the success of their actions.

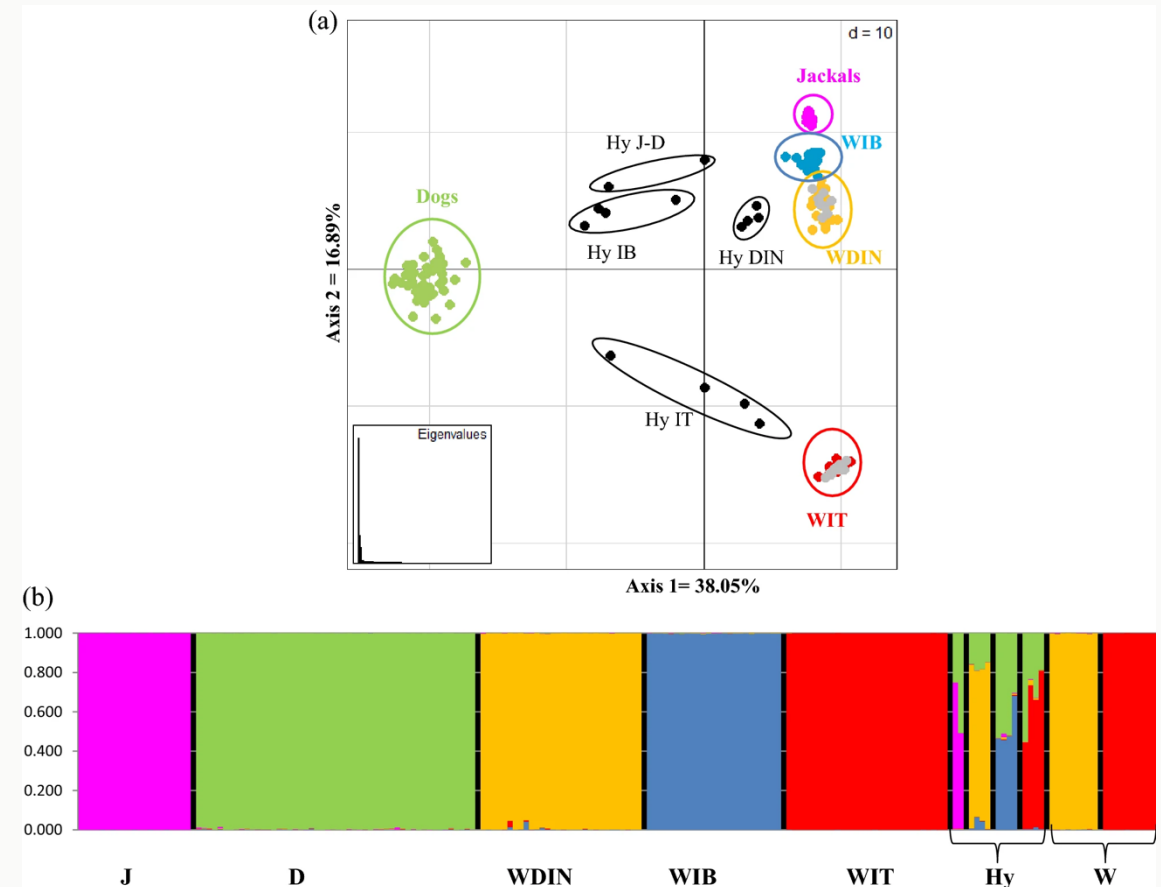
One species, multiple panels and goals



OPEN

A reduced SNP panel to trace gene flow across southern European wolf populations and detect hybridization with other *Canis* taxa

Astrid Vik Stronen^{1,2,3,13}, Federica Mattucci^{4,13}, Elena Fabbri⁵, Marco Galaverni⁵, Bernardino Cocchiara^{6,7}, Carsten Nowak^{8,7}, Raquel Godinho^{8,9}, Aritz Ruiz-González¹⁰, Josip Kusak¹¹, Tomaž Skrbinšek¹, Ettore Randi³, Albena Vlasheva¹², Nadia Mucci⁴ & Romolo Caniglia^{4,13}



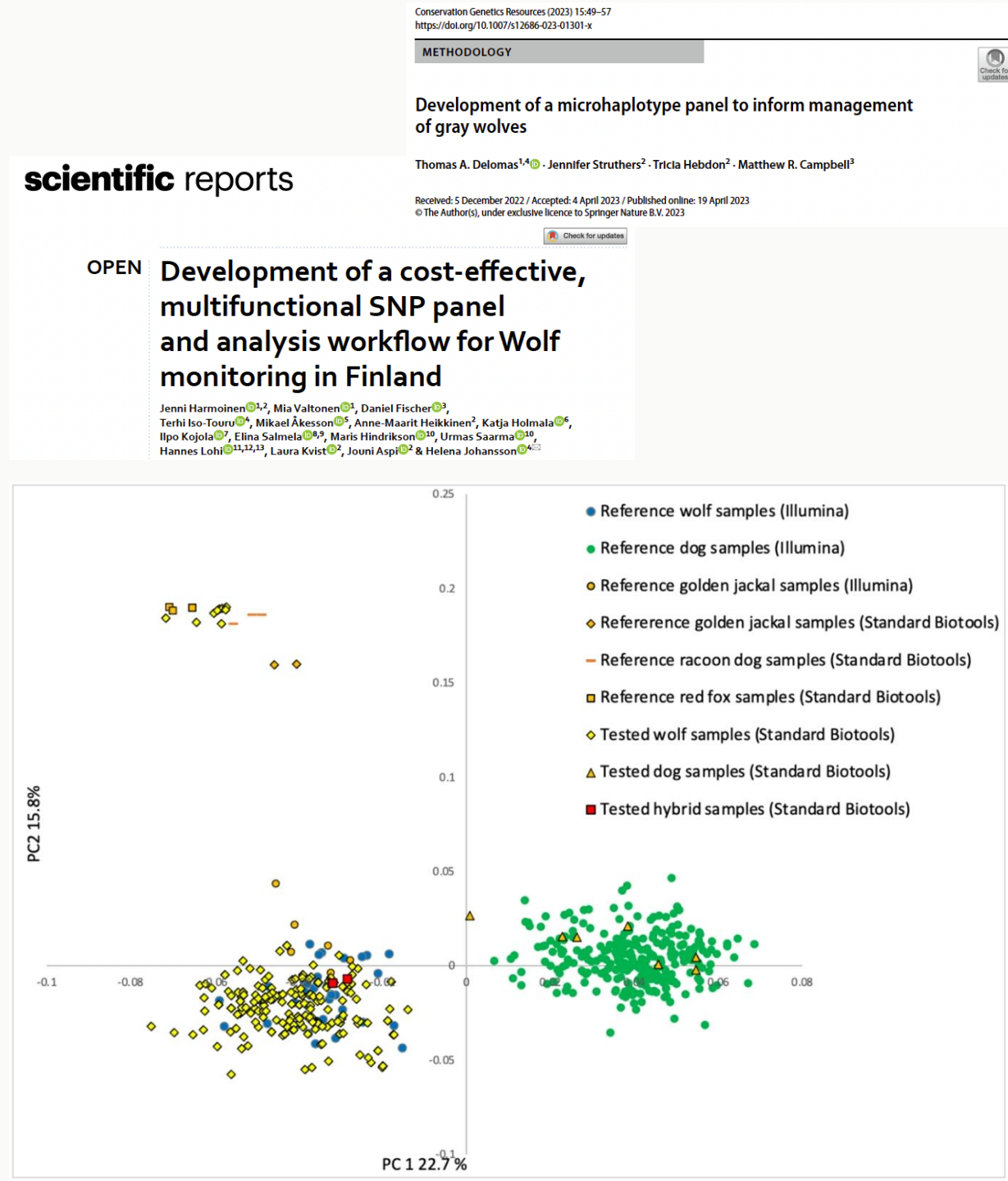
One species, multiple panels and goals

- Gray wolf recovery in North America and Europe must balance a recolonising carnivore against stakeholder conflict, and needs to reliably distinguish wolves from coyotes, dogs and hybrids — often from non-invasive samples like scat and hair.

One species, multiple panels and goals

- Gray wolf recovery in North America and Europe must balance a recolonising carnivore against stakeholder conflict, and needs to reliably distinguish wolves from coyotes, dogs and hybrids — often from non-invasive samples like scat and hair.
- Amplicon SNP / microhaplotype panels combine several jobs in one assay: a 341-locus panel for Idaho gray wolves adds individual ID and kinship inference (from 733 individuals, mean heterozygosity 0.57) alongside loci that specifically separate wolves from coyotes and from domestic dogs, plus a sex marker. A parallel 96-SNP panel for Finland is built directly for non-invasive genotyping from field-collected scat.
- Supports routine population monitoring, hybridisation tracking, and species/individual identification from field samples — without the cost or invasiveness of whole-genome sequencing.

Development of a custom-designed, multifunction SNP panel. It outlines the strategy to identify wolf genotypes from that of other species, sex determination, individual identification, and kinship analysis.



SNPs are one marker type among several

"SNP panel" is one answer to a broader question — which kind of genetic marker to genotype. Conservation genetics has used several:

mtDNA	Maternally inherited, non-recombining. Strong for species ID and phylogeography — but only one effective locus, no paternal signal.
Microsatellites (STRs)	Short tandem repeats scored by fragment length. Long-standing standard for individual ID, parentage and kinship — highly polymorphic per locus.
AFLPs / RFLPs	Older dominant-marker methods, largely superseded. Cheap in their day, but poor reproducibility and no allele-level resolution.
SNPs	Single base-pair variants, biallelic. Now the the most common marker class — scalable, standardised, and sequencer-native.
Microhaplotypes	Clusters of linked SNPs read together as a haplotype — SNP-level robustness with closer-to-STR information content per locus.

STR or SNP or others? Neither is 'better', they trade off differently

Property	Microsatellites (STRs)	SNPs
Alleles per locus	Many (often 5–20+) — high information per locus	Usually 2 (biallelic) — low information per locus
Loci needed for power	Fewer: ~10–20 for individual ID/parentage	More: ~50–500+ for the same statistical power
Mutation rate	High — useful for fine-scale kinship, but raises homoplasy risk	Low — high reproducibility, minimal homoplasy
Genotyping method	Fragment-length calling (capillary electrophoresis)	Sequence/allele calling (NGS, qPCR) — automatable
Error & reproducibility	Stutter, allelic dropout, null alleles; calling drifts across labs/platforms	Very low error rates; genotypes are directly comparable across labs and time
Degraded/non-invasive DNA	Works, but longer amplicons are more error-prone at low DNA quantity	Short amplicons perform well on scat, hair, museum material
Scalability / automation	Limited — hard to multiplex into large, high-throughput panels	Scales cheaply to hundreds–thousands of loci via NGS
Legacy data	Many established databases and decades of published baselines	Newer datasets — often not directly comparable to old STR baselines

Why the field has shifted toward SNP panels

SNPs trade information-per-locus for reproducibility, automation and comparability.

Conservation programmes increasingly need results that are standardised across years, labs and even species panels — exactly what STR fragment-length calling struggles to deliver at scale.

- Scalability: NGS makes 100s–1000s of SNPs cheaper per locus than a 15-locus STR panel run on a capillary sequencer
- Non-invasive samples: short SNP amplicons tolerate degraded DNA (scat, hair, museum skins) far better than longer STR amplicons
- Cross-lab standardisation: a SNP genotype is a discrete, sequence-based call — directly portable between labs, unlike lab-specific STR allele ladders

STRs still win when: a legacy dataset must stay comparable, or a very small, highly polymorphic panel (e.g. 10-15 loci) is genuinely enough for the task.

Why the field has shifted toward SNP panels

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TABLE 2 Cost per sample for 7 STR markers (Biello et al., 2021; Perez et al., 2014) and 41 SNPs (present work) for an increasing number of samples

Number of samples	7 STRs (in-house PCR, outsourced fragment analysis)	7 STRs (outsourced PCR and fragment analysis)	41 SNP panel (outsourced KASP, LGC genomics)
12	23.3€	34.4€	–
24	23.3€	34.4€	14.2€
48	23.3€	34.4€	10.6€
96	18.8€	30.1€	9.4€
192	18.8€	30.1€	7.0€

Note: STRs: fragment analysis quotation with or without PCR amplification (second and third columns, respectively) by Macrogen Europe; KASP quotation by LGC Genomics, UK (October 2021). For KASP, a minimum of 22 samples is required to determine the genotype of all the data points by means of genotyping cluster generation.

2. From Whole-Genome Sequencing to Targeted SNP Panels

- Why applied conservation work outgrows WGS
- The discovery → design → application workflow

WGS is unmatched for discovery — not for routine application

Whole-Genome Sequencing

- Genome-wide resolution — captures every variant class (SNPs, indels, SVs)
- Ideal for discovery, reference genome building, and fine-scale ROH/inbreeding estimates
- High per-sample cost at scale
- Requires high-quality, high-quantity DNA
- Computationally heavy; slower turnaround

Applied Conservation Needs

- Hundreds to thousands of samples, often repeated across years
- Degraded, low-quantity, non-invasive DNA (scat, hair, feathers, museum skins)
- Fast turnaround to inform real management decisions
- Standardised, comparable results across years, labs and platforms
- Budget that must scale with monitoring effort, not shrink it

Discovery, design, and application are separate phases

01

DISCOVERY

Few individuals

WGS / RADseq / WES across representative populations → variant calling → candidate SNP catalogue (10^5 – 10^7 SNPs)



02

PANEL DESIGN

Filtering & validation

Technical + biological filtering, purpose-driven selection, validation genotyping → a fixed marker set (10^1 – 10^3 SNPs)



03

APPLICATION

Many individuals

Targeted genotyping of large sample sets, repeated over time → parentage, assignment, monitoring answers

Genotyping platforms for a targeted panel

Platform	Typical panel size	Cost per sample	DNA quality tolerance	Best use case
Amplicon seq. (GT-seq, rapture)	50–500 SNPs	Low–moderate	High (degraded OK)	Large-scale routine monitoring, non-invasive samples
SNP arrays / chips	10^2 – 10^5 SNPs	Low at scale, high upfront	Moderate	Very large programmes, well-funded / model-adjacent species
KASP / TaqMan (qPCR)	1–50 SNPs	Very low	High	Small diagnostic panels, field-deployable assays
Microhaplotypes	50–300 loci	Low–moderate	High	Extra power per assay via linked-SNP haplotypes

All are typically preceded by a WGS/RAD-based discovery step

3. Criteria for SNP Selection

- Technical / assay-design constraints
- Biological and statistical filters
- Methods to select SNPs
- Panel size vs. information content

Constraints imposed by the assay itself

Clean flanking sequence

No repeats, paralogous regions or low-complexity DNA within ~100 bp — needed for specific primer/probe design and unambiguous mapping.

No neighbouring variants

Avoid SNPs or indels inside the primer/probe-binding region — a second variant there causes assay dropout or null alleles.

Sufficient depth & genotype quality

Retain only SNPs called with high confidence (adequate read depth, quality score) in the discovery data.

Biallelic

Most targeted platforms (KASP, arrays, most amplicon pipelines) are built for two-allele SNPs, not multiallelic sites.

Independence / spacing

Avoid strong physical linkage between panel SNPs unless haplotype information is deliberately wanted.

Cross-platform consistency

If several labs will genotype over years, favour SNPs that perform reliably across chemistries and instruments.

Filters that make a SNP informative, not just assayable

Minor allele frequency (MAF)

High enough across ALL target populations, not just the discovery sample — low-MAF or population-specific SNPs are the first entry point for ascertainment bias (→ Section 4).

Expected heterozygosity

Prefer SNPs with high polymorphism / informativeness for the intended task (identity, parentage, structure).

Hardy–Weinberg equilibrium

Strong deviations can signal null alleles, paralogs or genotyping error in the discovery data — flag or exclude.

Low missingness

Exclude SNPs poorly genotyped in the discovery panel; they will likely fail in the field, too.

Neutrality (usually)

For diversity/structure panels, exclude loci under strong selection. For adaptive-monitoring panels, this criterion is deliberately reversed.

There is no single 'best' SNP — only the right SNP for the task

Individual ID & Parentage

~100–200 SNPs

Highly polymorphic, unlinked, neutral SNPs with high power for exclusion and kinship.

Population Assignment

~50–300 SNPs

F_{st}-outlier / highly differentiated SNPs between reference populations — ancestry-informative markers (AIMs).

Adaptive / Functional Monitoring

Tens–hundreds

SNPs linked to genes under selection (climate adaptation, disease resistance) — deliberately non-neutral.

In practice, not all operational panels blend all three categories into one multi-purpose marker set.

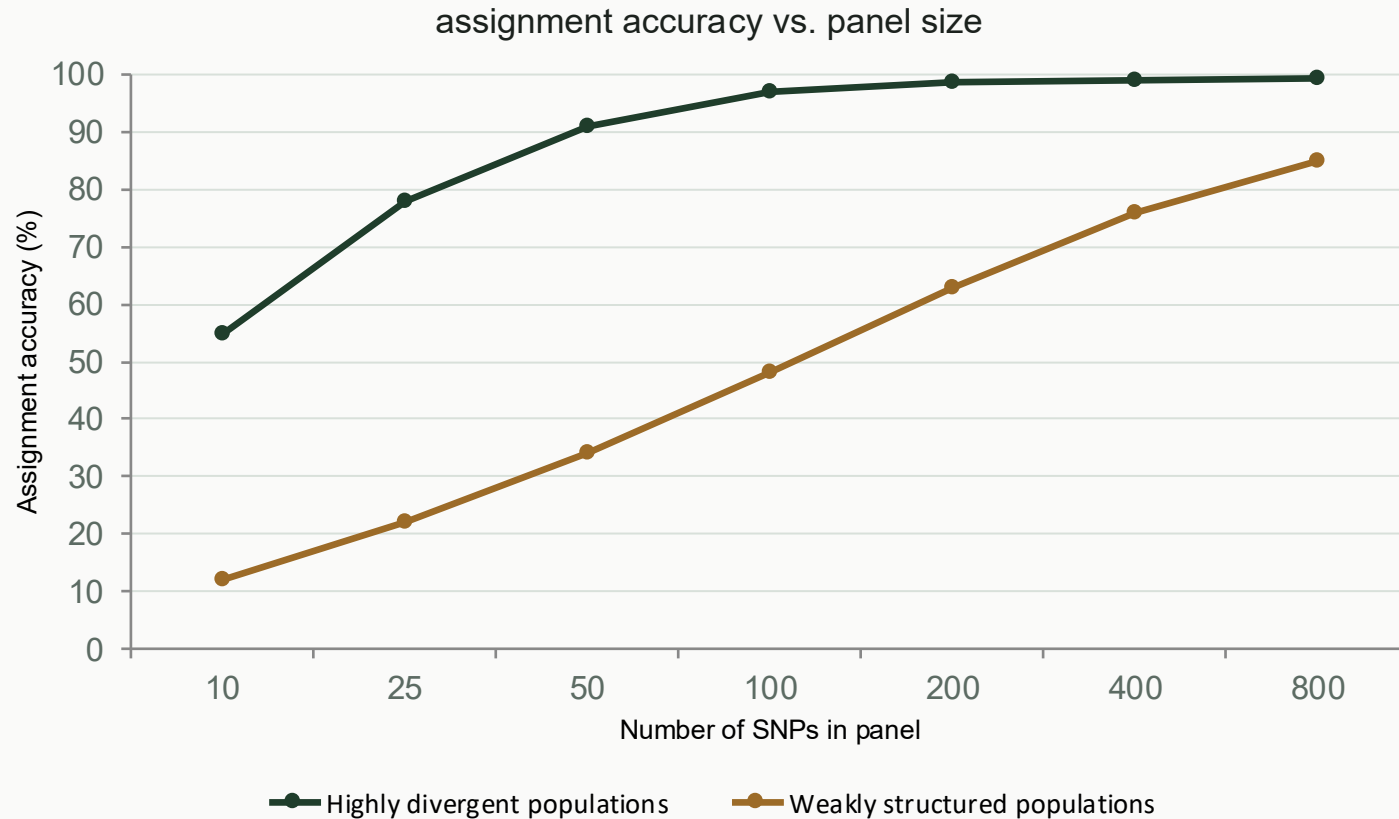
How candidate SNPs are actually picked out

The criteria on previous slides say what a good SNP looks like. These are the statistical and computational methods used to find those SNPs among thousands–millions of candidates.

Method	What it flags	Typical panel use	Example tools
Fst outlier scans	SNPs far more differentiated between populations than the neutral genome-wide background	Population-diagnostic / assignment (AIM) panels	BayeScan, OutFLANK, PCAdapt
PCA (loadings)	SNPs driving the main axes of genetic structure in the discovery data	Confirming/refining structure-informative SNPs	PCA loadings, PCAdapt
DAPC	SNPs that best discriminate predefined groups (populations, species, hybrid classes)	Group-diagnostic and hybrid-ID panels	adegenet (R)
Random Forest / ML importance	SNPs with the highest classification importance, considered jointly rather than one at a time	Robust multi-purpose or complex-structure panels	randomForest, ranger (R); scikit-learn, AssignPop
Genotype–environment association (GEA)	SNPs whose frequency correlates with an environmental gradient	Adaptive / functional monitoring panels	RDA, LFMM, BayEnv
LD pruning	SNPs that are statistically redundant with a neighbour	Enforcing independence in the final panel	PLINK, SNPRelate

These methods narrow the candidate pool down to the panel-sized shortlist — which then still has to pass the technical checks and validation.

More SNPs help... until they don't



- Returns diminish quickly once a panel captures the dominant axis of differentiation.
- Divergence between source populations (F_{st}) sets the ceiling — more SNPs cannot fully compensate for weak structure.
- Curves shown are illustrative, not empirical — real curves depend on the species, markers and reference baseline.

4. Assignment Tests

- What an assignment test does
- How assignment is computed
- Testing panel power: self-assignment
- Thresholds

Inferring where an individual came from

A statistical method that assigns an individual of unknown origin to its most likely population, using a reference baseline of genotyped individuals from known populations.

Wildlife forensics

Tracing poached / trafficked
wildlife products to their
population or region of origin

Provenance verification

Confirming the claimed source
of fisheries, timber or other
regulated products

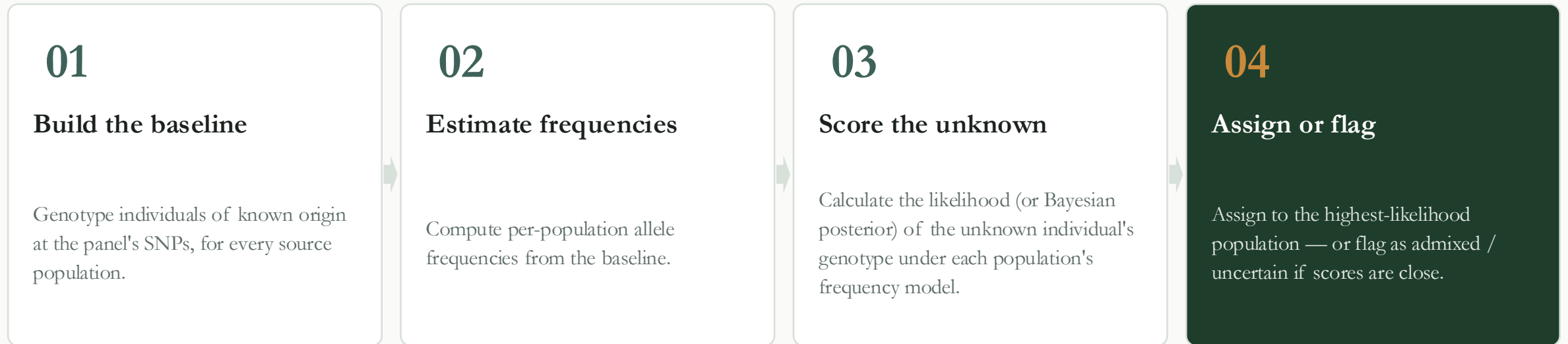
Migrants & hybrids

Detecting individuals that
moved between populations, or
admixed offspring

Captive-breeding & reintroduction

Verifying the genetic origin of
founders and released
individuals

From reference baseline to an individual assignment



Common approaches:

Bayesian / frequentist likelihood (e.g. Rannala–Mountain, Paetkau methods) • clustering-based (STRUCTURE, ADMIXTURE) • multivariate (DAPC) • machine learning (e.g. Random Forest)

Self-assignment: does the panel actually work?

Leave-one-out cross-validation

- Remove one individual from the reference baseline, reassign it using everyone else, and check whether it comes back to its true population.
- Repeat for every baseline individual → report assignment accuracy (%) per population.

Also simulate unknowns at varying admixture levels to test the panel's power to detect hybrids and recent migrants specifically.

Why this matters

- Confirms the panel is fit for purpose before it goes into a forensic or management pipeline
- Exposes populations the panel struggles to separate — informs whether more/different SNPs are needed
- Provides the accuracy figure you must report alongside any real-world assignment result

How strict should an assignment threshold be?

Every assignment is a probability. The threshold decides when we act on it, and which mistake we would rather make.

more calls, more errors

RELAXED

e.g. most likely source, $q \geq 0.8$

Be more relaxed when...

- It is a first screening, confirmed later with more markers
- You track proportions or trends at population level
- The action is reversible (extra sampling, temporary holding)
- Missing a case is the worse error (e.g. flagging fish from a protected run)

fewer errors, more "unassigned"

STRICT

e.g. $q \geq 0.95-0.99$

Be strict when...

- The result goes to court (wildlife forensics)
- Animals will be released into the wild (translocation)
- The action is irreversible (e.g. lethal removal of hybrids)
- A wrong call could harm a small, threatened population

Rule of thumb: set the threshold by the cost of being wrong, and report the self-assignment accuracy at that threshold.

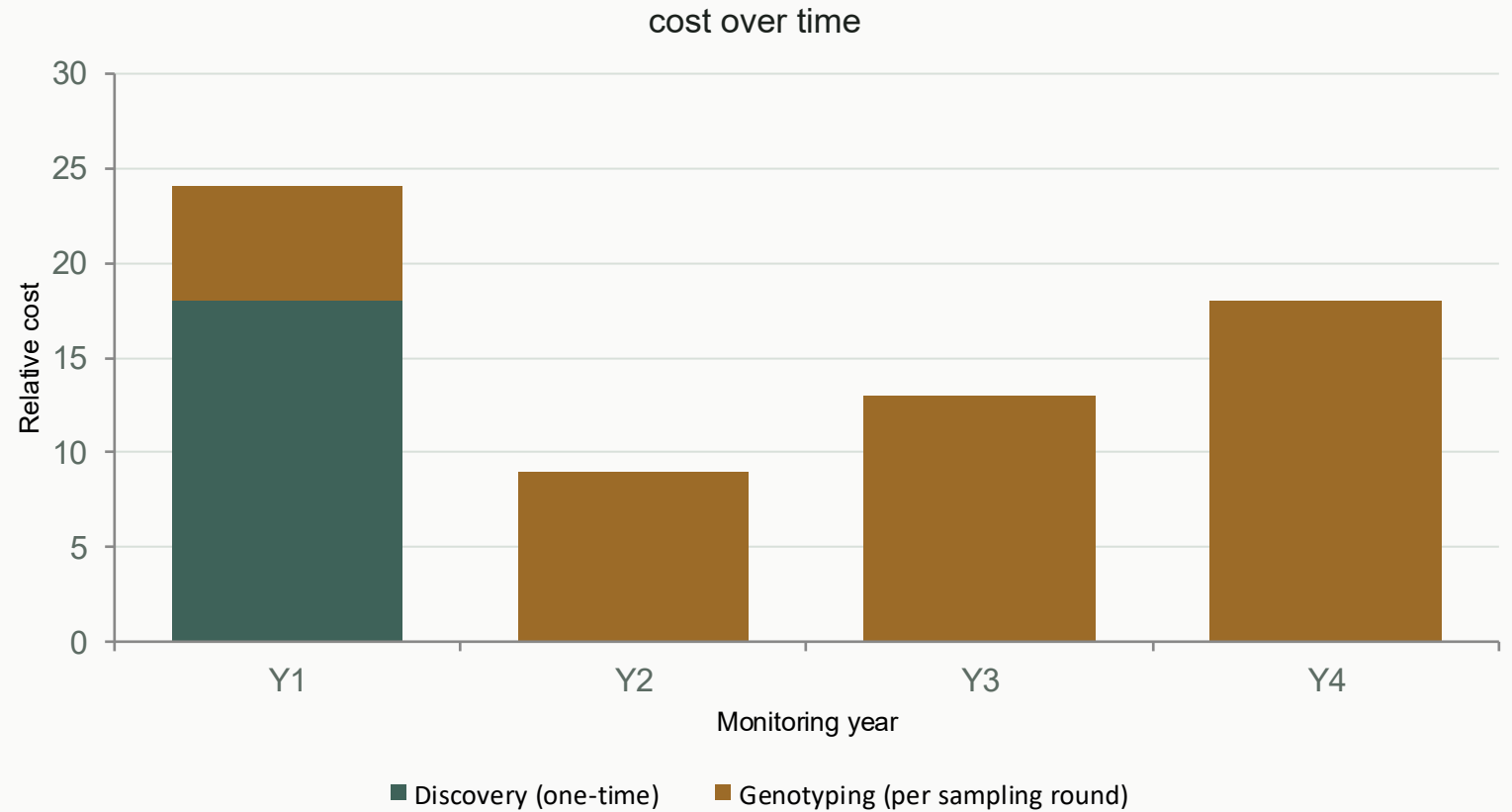
5. Cost-Efficiency & Scalability

- The two-phase cost structure
- Comparing platforms on cost & throughput
- Scalability beyond raw cost
- A decision framework: WGS vs. panel

A monitoring programme has two very different cost phases

One-time, fixed cost

A moderate number of individuals,
sequenced deeply (WGS/RAD).
High cost per sample — but paid once.



As sampling rounds accumulate, recurring genotyping cost — not the one-off discovery cost — dominates total programme spend.

Genotyping techniques used to run a panel

Technique	Example products	Cost / sample	Throughput	DNA tolerance
SNP array / chip	Illumina Infinium, Affymetrix Axiom	Low at scale, high upfront	Very high (fixed panel)	Moderate
KASP (allele-specific PCR)	LGC KASP	Very low	High (dozens of markers)	Tolerant
TaqMan (qPCR)	Thermo Fisher TaqMan	Very low	High (dozens of markers)	Tolerant
Amplicon sequencing (GT-seq)	GT-seq (Campbell et al.)	Low–moderate	High (multiplexed NGS)	Tolerates degraded DNA
Multiplex amplicon panels	MonsterPlex, PlexSeq, Rapture	Very low per target at scale	Very high (1000s of samples pooled)	Works from ~20–40 ng DNA
Microfluidic PCR arrays	Fluidigm / Standard BioTools	Moderate	Moderate (100s of SNPs × samples)	Moderate

Excludes WGS and RADseq, those are discovery-phase methods, not ways of running an already-designed panel.

Scalability is more than cost per sample

- **Automation & multiplexing** 96/384-well robotics and barcoded pooling for sequencing determine real throughput ceilings.
- **Required turnaround time** Real-time anti-poaching response vs. an annual monitoring report call for very different platforms.
- **Field sample type** Non-invasive DNA (scat, hair, feathers, eDNA) and museum material favour amplicon/qPCR platforms over WGS.
- **Long-term reproducibility** A fixed, validated panel gives comparable data across years and labs; repeated re-sequencing risks batch effects.
- **End-user bioinformatics capacity** Management agencies and field teams need simpler outputs than a research lab's re-sequencing pipeline.

So, WGS or a targeted panel?

Stay with WGS when...

- Building the initial reference / discovery resource
- You need the full allele spectrum (e.g. fine-scale ROH, adaptive loci discovery)
- Sample numbers are small
- Budget and turnaround allow it

Move to a targeted panel when...

- Monitoring is routine and repeated over years
- Sample numbers are large (hundreds–thousands)
- Samples are degraded or non-invasive
- Decisions need fast turnaround, and results must stay comparable across time and labs

Summary

- SNP panels already underpin real conservation decisions — fisheries management, wildlife forensics, carnivore monitoring — not just research papers.
- SNP panels trade genome-wide resolution for cost, speed and scalability — WGS remains essential for discovery.
- Panel design balances technical assay constraints with biological purpose (ID/parentage, assignment, adaptive monitoring).
- Assignment power depends on population divergence, reference-baseline quality, and SNPs ascertained for that specific task.
- Platform choice should be driven by total programme cost over time, not just per-marker discovery cost.

Discussion questions

- A management agency needs real-time hybrid detection in the field within days of sample collection. Which genotyping platform would you recommend, and why?
- How would you validate that a 96-SNP panel, designed from 10 WGS individuals, is trustworthy before deploying it to genotype 2,000 field samples?
- Would you prioritise a small, highly-informative assignment panel or a broader neutral-diversity panel for a species with only weak population structure? Why?
- Your discovery samples come from only the northern edge of a species' range. What specific ascertainment biases would you expect, and how would you test for them?

Ascertainment Bias

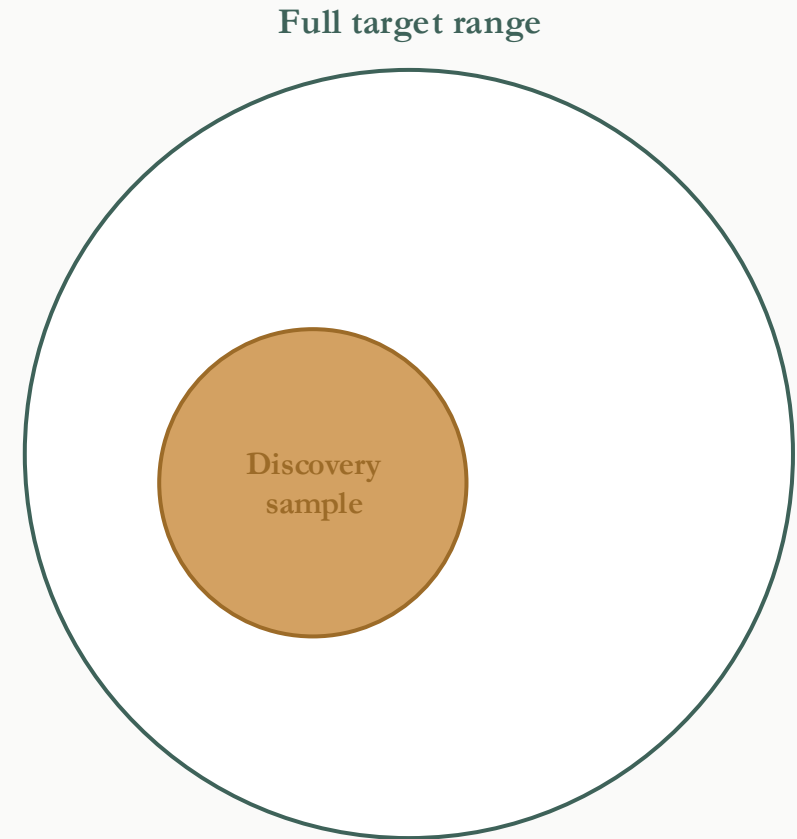
- What ascertainment bias is
- Where it enters the pipeline
- What it distorts downstream
- How to limit and report it

A panel reflects the sample it was discovered in

Ascertainment bias:

the systematic bias introduced when the individuals used to DISCOVER SNPs differ from the population(s) the panel is later APPLIED to. The resulting markers over-represent variation that is common in the discovery sample.

It is not a bug in one dataset — it is a structural property of any panel built from a limited discovery sample.



Panel SNPs are ascertained toward whatever lies inside the small circle — and are less informative, or misleading, outside it.

Four common entry points

Small / unrepresentative discovery sample

Too few individuals, or individuals drawn from only one or two accessible populations.

Reference genome bias

SNPs are called relative to one individual's genome — variation in more divergent lineages is under-called or mismapped, inflating apparent missingness or diversity loss precisely where the reference is least representative.

MAF / frequency filtering

Removing rare or private alleles during panel design disproportionately strips out variation specific to small or genetically distinct populations.

Cross-platform compatibility filtering

Narrowing candidates further to SNPs that work on every intended chemistry/lab shrinks the pool again, often toward the best-characterised samples.

What a biased panel distorts

Underestimated diversity

H_e and π appear artificially low in populations far from the discovery sample.

Distorted allele frequency spectra

Biases demographic inference (e.g. bottleneck / expansion signals).

Skewed F_{st} / differentiation

Estimates can be inflated or deflated depending on the direction of the bias.

Phylogenetic 'attraction'

Populations related to the discovery sample cluster artificially close together.

Reduced power in under-sampled groups

Weaker ability to detect real structure or unique variation there.

You can't eliminate ascertainment bias — but you can manage it

- Use a large, geographically and genetically representative discovery panel spanning the full target range
- Consider multiple reference genomes or a pangenome approach where divergence is high
- Report discovery-panel composition transparently in methods — this is a reviewer's first question
- Apply ascertainment-aware models downstream where representative discovery isn't feasible, and interpret cross-population diversity comparisons cautiously
- Re-validate and update the panel as new populations are sampled over time

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